

When Do Agent Harness Improvements Generalize? Semantic Binding, External Validity, and Evaluation Reliability in LLM Agents

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Abstract

A Harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families. On a controlled tool-grounded *semantic-handoff* task, a non-answer-bearing “clarity bundle” (BundleS) suppresses a specific tool-green semantic-binding failure for one model, confirmed on a pre-frozen held-out instance (Study I). But the effect does not transfer: under exact counterbalancing it is not established for a second model, and on two structurally independent task families it is a clean null or a non-discriminative ceiling (Study II). Expanding an external STA evaluation prospectively from three to twelve independently frozen task instances reversed the apparent direction of the comparison while still failing to establish a stable Harness benefit. We contribute a *Harness Effect Audit Protocol*—four layers (capability, sampling, execution, artifact integrity) instantiated and stress-tested in this study—each shown catching a real threat; an independent benchmark-repair audit (Terminal-Bench 2.0→2.1) shows broad but predominantly partial correspondence (22 of 26 repairs had at least partial correspondence, only one directly captured). Our controlled task instances test existence, recurrence, and transfer direction rather than estimate population-level pass rates. A preregistered blinded human construct-validity study is reported as unexecuted. The contribution is an evaluation study, not a proposed improved harness.

1 Introduction

A harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families. This is the result that motivates the paper: an effect that looks robust under repeated within-condition replication and a frozen held-out instance can still fail the first strict external-validity test—and, as we show, can even reverse direction when an external family is expanded prospectively. If the community treats a single within-family replication as evidence that a harness improvement “works,” it will over-credit interventions that are model- or family-specific.

We study this on tool-grounded EDA workflows, where an LLM agent binds quantities to canonical typed roles and submits an artifact to a real commercial tool. On a *semantic-handoff* task the agent must assign a tuple to canonical typed roles from heterogeneous, role-misleading evidence; a wrong binding is not a syntax error—it produces a *plausible green signoff*—so only a typed provenance/authority grader distinguishes correct from wrong. We ask: **RQ1** can harness information structure change semantic-binding behavior (controlled setting)? **RQ2** does it transfer across models and task families? **RQ3** which evaluation controls make these conclusions valid?

Contributions. (1) A controlled semantic-binding mechanism + typed failure taxonomy, isolated from analog-design difficulty (Study I). (2) The transfer-failure-after-held-out-confirmation finding, including a prospective 12-instance STA expansion that reversed the apparent direction of the original three-instance pilot while still not establishing a stable benefit (Study II). (3) The Harness Effect Audit Protocol, instantiated and stress-tested here, plus an independent benchmark-repair audit as external taxonomy application (Study III).

2 Related Work and Problem Formulation

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request-authority join. Families A/B are structurally independent from the workflow family (verified under five pre-registered criteria; operational, not a proof).

Positioning. Prior work establishes that harness configurations affect agent performance. We instead study *when* a measured effect is scientifically attributable, *whether* it survives external-validity tests, and *which* controls are required (Table 1). Two further distinctions: **TAB (No More, No Less: Task Alignment in Terminal Agents)** [2] studies the selective use of necessary environmental cues in the presence of plausible distractors; our construct is role-conditioned semantic binding and provenance/authority attribution, including cases where the evidence is available but values are bound to the wrong semantic role while the tool stays green—a different failure construct. **LongHorizon-Harness** [1] reports cross-benchmark Harness gains; rather than contradicting us, this sharpens our claim: *some* Harness interventions do transfer across benchmarks, so transfer is an empirical property that must be demonstrated rather than inferred from within-family confirmation.

Table 1: Positioning relative to five lines of related work (paraphrased; citations are VERIFICATION-REQUIRED placeholders).

Prior work	What it establishes	What this paper additionally does
Harness-Bench	Broad model×harness characterization	Controlled semantic-mechanism isolation + typed taxonomy; claim-validity analysis
Rethinking Harness Evaluation	Leaderboard search/overfitting	Held-out confirmation still need not transfer cross-family
HarnessOpt-Bench	Optimizer capability	Studies validity of harness-effect claims, not optimization
ABC / Best-Practices	Setup; shortcut/reward validity	Additionally audits transport, membership, tool health, action surfaces, canonical integrity
Protocol-validity work	Protocol design; reward-hacking surface	Measurement validity as the contribution; per-control threat log

3 Harness Effect Audit Methodology

Tasks and tool grounding. All families run on real commercial tools (PrimeTime; HSPICE) through a transparent remote shim—not tool simulations. A wrong binding produces a plausible green signoff/number, so tool success is necessary but not sufficient; only the typed provenance/authority grader rejects a wrong binding.

Conditions. **Base** (ambiguous handoff; no answer-bearing disclosure); **BundleS** (canonical labels C1, value-domain definitions C2, glossary C4, procedural contract C7; no answer-bearing C6; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions.

Design. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Task instance is the primary unit; stochastic repetitions are nested. We report paired directions and raw counts per instance and do not pool trajectories into a single significance headline.

Detectability. With small instance counts our design is diagnostic rather than population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes. We do not use small- n bootstrap significance or a trajectory-pooled p -value to compensate.

Audit protocol (preview; Figure 1). Four layers, each reported independently: (1) capability validity, (2) sampling validity, (3) execution validity, (4) artifact integrity. We describe the protocol as instantiated and stress-tested in this study—not broadly validated or universally reusable.

L1 Capability	tool-green alternatives; provenance/authority oracle; semantic failure taxonomy
L2 Sampling	repeated trajectories; blocking/counterbalancing; committed membership arbiter
L3 Execution	streaming transport; terminal-vs-recovered; tool-health bookends; protocol completion
L4 Artifact	action-surface isolation; exact-commit worktree + frozen hashes; custody + sanitized logs

Figure 1: The Harness Effect Audit Protocol (four layers, nine controls).

Experimental status. Table 2 records the registration type of each result; only predeclared, frozen-held-out, exact-counterbalanced, and prospective-confirmatory results are headline. This directly answers the sequential-adaptation concern.

Table 2: Experimental-status registration.

Registration type	Result (used in)	Headline?
Predeclared (controlled pair)	Workflow family: clarity bundle suppresses axis binding (Study I)	Yes
Pre-frozen held-out	Workflow family: BundleS on a pre-frozen held-out truth (Study I)	Yes
Exact-counterbalanced	Cross-model: BundleS vs Base, balanced blocking (Study II)	Yes
Prospective confirmatory	STA external family: 12 pre-frozen new instances (Study II)	Yes
Secondary extension (pre-frozen)	TypedContract; SPICE ceiling (Study II)	Yes (scoped)
Exploratory (sequential)	Component localization C1/C2/C4/C7/C24 (Appendix)	No
Preregistered but unexecuted	Blinded human construct-validity study (Study B)	n/a

4 Study I — Controlled Semantic-Binding Effects (RQ1)

Question: can harness information structure change semantic-binding behavior, holding truth and grader fixed? On the workflow controlled pair (identical truth/grader; ambiguous vs. clear handoff), the full clarity bundle suppresses the axis-binding failure for Model A: Base 1/3 (2 axis-binding) → BundleS 3/3 (0 axis-binding). The schema/contract subset BundleS (excluding the answer-bearing C6) suffices on the development pair, and generalizes to a *pre-frozen held-out* instance: BundleS 3/3 vs Base 1/3. The failure the bundle suppresses (Figure 2) partitions the workflow ledger into 29 correct, 20 *axis-binding* failures, and 5 *role-conditioned value-selection* failures. **RQ1 is supported within the controlled workflow family for Model A; we make no universal improvement claim.**

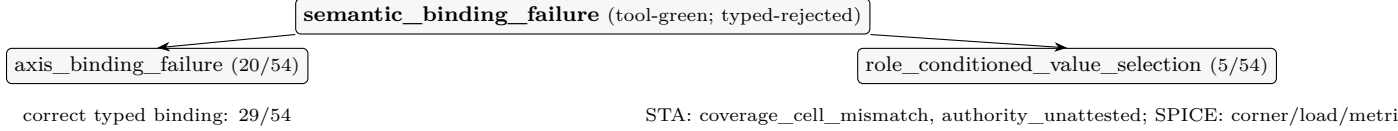


Figure 2: Semantic-binding failure taxonomy.

5 Study II — Model and Task-Family Transfer (RQ2)

Question: does the within-family effect transfer across models and families? Table 3 is the main result. The RQ2 headline separates three outcomes: *null/unestablished transfer* (STA, discriminating; Model B cross-model), *ceiling* (SPICE, non-discriminative), and *direction reversal under prospective expansion* (STA pilot vs. prospective).

Table 3: Main result matrix (semantic-binding correct rate; instance is the unit).

Family / model	Base	BundleS	TypedContract	n	verdict
Workflow / Model A (development)	1/3	3/3	—	1	controlled effect
Workflow / Model A (frozen held-out)	1/3	3/3	—	1	held-out confirmed
Workflow / Model B (exact-counterbalanced)	3/4	3/4	—	1	no benefit
STA / Model A (prospective)	0.208	0.333	0.458	12	transfer not established
SPICE / Model A	1.00	1.00	1.00	3	ceiling (non-discriminative)

STA—prospective confirmatory batch (authoritative). Across 12 pre-registered new instances, BundleS vs Base produced 3 improving, 2 declining, and 7 tied instances; the exact sign-test $p = 1.0$ and the pre-registered instance-level permutation sensitivity $p = 0.31$. *Prospective STA confirmation did not establish a transferable BundleS benefit. The descriptive direction reversed relative to the original three-instance pilot, illustrating the instability of conclusions drawn from very small task samples.* We do not claim a significant BundleS (or TypedContract) improvement, nor a stable STA negative. TypedContract is descriptively highest but not established. The historical three-instance STA pilot (Base 0.50 / BundleS 0.33 / TypedContract 0.33) is reported separately as descriptive pilot evidence and is **not pooled** with the prospective $n = 12$.

Why the direction reversal matters. The pilot (Base 0.50 $\geq$ BundleS 0.33) and the prospective batch (Base 0.208, BundleS 0.333) disagree on the *sign* of the comparison: under the pilot one would have written “BundleS slightly hurts STA”; under the prospective batch one would write “BundleS slightly helps STA.” Neither is statistically distinguishable from no effect. The reversal is therefore not evidence of a real change in mechanism—it is a direct demonstration that three task instances cannot pin even the direction of a harness comparison on this family. Per-instance, the prospective task is hard: many instances sit at the floor across all three conditions, and the condition differences are concentrated in a handful of discriminating instances; within-instance two-repetition agreement is high (only one of twelve Base cells disagrees with itself across repetitions), so between-instance spread, not run-to-run noise, drives the family means.

Cross-model + SPICE. Under exact counterbalancing the within-family BundleS benefit is not established for Model B (3/4 = 3/4). Family B (SPICE) is at ceiling (1.00 across conditions)—non-discriminative, not a refutation.

RQ2. *Within-family held-out confirmation did not establish cross-model or cross-family transfer.*

6 Study III — Evaluation Validity and Independent Benchmark Repair Audit (RQ3)

Each headline conclusion could have been manufactured or masked by an evaluation artifact. The four-layer protocol is instantiated and stress-tested in this study; each control family caught a concrete threat (full episode log in the Appendix):

- **L1 (capability):** a wrong-axis binding that scored tool-green, visible only to the typed grader—without L1 the very failure the bundle suppresses would have been scored correct.
- **L2 (sampling):** the same controlled-pair instance is non-deterministic across trajectories; repetition + exact counterbalancing + a committed membership arbiter make the directional claim defensible and prevent a transport-invalid episode from being scored as a capability failure.
- **L3 (execution):** a non-streaming HTTP transport censored a thinking model’s long reasoning and was misread as a capability failure; SSE streaming recovered the real behavior. (Pre-flight it also caught a grading-path bug—a malformed signoff-status JSON that would have forced a component false for every STA episode—which was fixed before the prospective run.)
- **L4 (artifact integrity):** a silently-rewritten canonical golden was misreported as a tool outage, and a SPICE action-surface rule manufactured false zero-scores; both were caught and the latter repaired before the prospective run.

Independent Benchmark Repair Audit (Terminal-Bench 2.0→2.1). As external taxonomy application, we coded the 26 officially documented repairs (PR #53 “Terminal-Bench 2.1” in the Terminal-Bench 2 repository) under the four-layer-plus-Other rubric, *frozen before inspecting the repairs*. This is a retrospective application of a pre-registered taxonomy to validity repairs made independently by another benchmark team. Result: **direct 1 / partial 21 / not-covered(Other) 4** of 26; layers capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer. *The taxonomy showed broad but predominantly partial correspondence with independently documented benchmark repairs: 22 of 26 repairs had at least partial correspondence, but only one was directly captured and four lay outside the taxonomy.* This is **not** runtime validation; it does not show our protocol predicted or would necessarily have prevented the repairs; and the absence of sampling-layer repairs means the external audit does not validate every protocol layer. The full 26-task coding + source evidence is in the Appendix.

7 Limitations and Discussion

Controlled executable tasks, not sampled production incidents—the tasks isolate the semantic-binding mechanism. *EDA-heavy environments*—the transfer result is established only within EDA. *Small instance counts*—the design is diagnostic, not population-estimating; the STA prospective null cannot rule out a small effect (sign $p = 1.0$, permutation $p = 0.31$), and the direction reversal between the $n = 3$ pilot and $n = 12$ prospective batch illustrates small-sample instability rather than estimating a rate. *Construct validity*—our semantic-binding labels rest on **executable provenance/authority oracles rather than independent blinded human judgment**: the preregistered human construct-validity study (Study B) is **preregistered but unexecuted because qualified independent human annotators were unavailable**; no LLM annotator was substituted, and we do not claim independent human validation of the taxonomy. *Model scope*—the within-family effect is established for one model; cross-family evaluation used that model. *Audit protocol*—instantiated and stress-tested here, not broadly validated; the Terminal-Bench audit is retrospective taxonomy application, not prediction/prevention. *No mechanism isolated*—the bundle-level effect is real within-family for Model A but no stable minimal component was identified (Appendix).

Discussion. The result is a caution and a method. The caution: a single within-family replication—even a frozen held-out one, and even a small external pilot—is weak evidence of generalization for harness effects. The method: the audit protocol makes the positive and null conclusions

independently checkable. Some Harness interventions do transfer across benchmarks; transfer is therefore an empirical property that must be demonstrated rather than inferred from within-family confirmation.

Reproducibility. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching; sanitized custody evidence. Frozen manifests, the treatment mapping, schedules, and provenance are in anonymized supplementary material. Program totals: 58 (workflow) + 24 (cross-family core) + 36 (cross-family secondary) + 72 (prospective STA) paid primary episodes; committed-ledger cost ¥745.29.

8 Conclusion

A harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families; prospectively expanding an external family from three to twelve instances reversed the apparent comparison direction while still not establishing a stable benefit. We contributed the Harness Effect Audit Protocol (four layers, stress-tested) and applied it retrospectively to 26 independent benchmark repairs (broad but predominantly partial correspondence). The contribution is an evaluation study—of *when* an effect is attributable, *whether* it transfers, and *which* controls make either claim credible—not a proposal of an improved harness.

References

- [1] LongHorizon-Harness authors. Longhorizon-harness: Advancing long-horizon agents for real-world tasks, 2026. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation. Reports cross-benchmark Harness gains.
- [2] TAB authors. No more, no less: Task alignment in terminal agents, 2026. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation. Studies selective use of necessary environmental cues amid plausible distractors.

Appendix outline (full content in the anonymous supplement)

A. Study I per-instance + failure-subtype tables; minimal-component (C1/C2/C4/C7/C24) exploratory decomposition. B. Study II per-instance prospective STA table (12 instances $\times$ 2 reps $\times$ 3 conditions; within-instance agreement); historical pilot (n=3) kept separate. C. Study III four-layer $\rightarrow$ nine-control operationalization; concrete caught threats; the full 26-task Terminal-Bench repair audit with exact source evidence. D. Operational independence criteria; historical repository phase chronology. E. Infrastructure implementation + custody hashes. (The committed supplement contains all episode evidence; this PDF’s appendix is finalized in the supplement.)

Ethics statement. The preregistered blinded human construct-validity study (Study B) is *preregistered but unexecuted because qualified independent human annotators were unavailable*; no LLM annotator was substituted. The manuscript does not claim independent human validation of the semantic-binding taxonomy; construct validity rests on executable provenance/authority oracles.

AI-use statement (outside the page limit). An LLM-based coding/writing assistant was used for orchestration code, generators, infrastructure, report scripts, and prose drafting. All scientific results are produced by committed deterministic pipelines reproduced from frozen manifests; no claim was authored or validated by an LLM as an oracle; all numbers were generated by committed code and re-extracted from committed ledgers.

Reproducibility statement. See above; frozen manifests, generators, graders, schedules, custody hashes, and sanitized episode evidence are in the anonymized supplement. Exact submission HEAD and custody hashes are recorded at freeze.